RICCARDO DI GIROLAMO

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Education

TU Delft, Master of Science in Mechanical Engineering — **Delft, Netherlands**, Sept 2022 – Dec 2024

- *Final Grade*: *9 cum Laude*
- *Relevant courses*: Mechatronic Systems Design, Precision Mechanisms Design, Robot Software Practicals (ROS2, C++/Python), Multi-Body Dynamics, Control Systems Design.

Politecnico di Milano, Bachelor of Science in Aerospace Engineering — **Milan, Italy**, Sept 2019 – Sept 2022

- *Final Grade*: *101/110* | *Merit scholarship*: *Most Outstanding Freshmen (2020)*.
- *Relevant courses*: Dynamics of Aerospace Systems, Structural Mechanics, Aerospace Propulsion, Flight Mechanics, Aerospace Materials.

Collegio di Milano, Resident Student & Student Representative — **Milan, Italy**, Sept 2019 – July 2022

- Selective student residence for high-achieving students offering seminars, workshops, and company training with a focus on digital transformation (GPA requirement 27/30).
- Elected Student Representative in the A.Y. 2021-2022.

Experience

Self-Initiated Project, Robotics/Mechatronics Engineer — End-to-end design of a ROS 2 differential-drive robot — **Delft, Netherlands**, May 2025 – July 2025

- Designed and prototyped a modular differential-drive robotic platform (parametric CAD in SolidWorks, aluminum frame + 3D-printed parts).
- Developed the embedded firmware (C++) and the ROS 2 Control integration via a custom HardwareInterface (C++/Python).
- Ran a structured test plan assessing: command-to-motion latency, max stable speed, payload, and power consumption.
- Produced clear build documentation, test procedures, demo video, and a public repository. (Project page).

TU Delft, Researcher – Master's Thesis — **Delft, Netherlands**, Sept 2023 – Dec 2024

- Built an optimization framework for multi-axis additive manufacturing that minimizes part deformation — C++/Python, PETSc (MPI). (Project page).

SymPy - Google Summer of Code (GSoC), Software Developer (Python) — **Remote**, May 2024 – Aug 2024

- Built a benchmark suite for *sympy.physics.mechanics* and *sympy.physics.vector* modules to analyze performance and track bottlenecks.
- Reimplemented *jacobian()* and *partial_velocities()* functions, achieving up to 2.7x and 24x faster execution times, respectively, on relevant benchmarks.
- Collaborated with maintainers to refine code quality, add tests, and improve documentation for long-term maintainability. (Project page).

Extracurricular Activities

- Participated in the LeRobot Worldwide Hackathon (June 2025, Eindhoven), experimenting with Visual-Language-to-Action (VLA) models for robotic control.
- Member of the FabLab where I learned and practiced with rapid prototyping and digital fabrication (3D printing, laser cutting, CNC milling, PCB fabrication, soldering). (L'Aquila, Italy - 2015-2019)

Skills

Hardware Design, Prototyping, and Testing: SolidWorks, Design for Manufacturing (DFM), PCB Design, FDM 3D Printing, Hand Tools, Soldering, Microcontrollers (Arduino, ESP32), BLDC Motors, Drivers, Encoders, Oscilloscope, Multimeter

Software and Programming Languages: ROS 2, ros2_control, Nav2 (basic familiarity), Gazebo, C++, Python, MATLAB, LaTeX, Git, Linux

Languages: Italian (native), English (C1)